Double slash comment & multi-line

- The program will skip any one of them.
- Usage of comments in your programming will facilitate its understanding.
- It can improve debugging and maintenance in future applications.
- The double slash (//) will allow you to comment on one line.
- Multi-line comment (/* */) will allow you to comment on multi-lines as long as you don't close it

//	This double slash allows you to write any sentence in the program because the program will ignore this line once it is detected. Mainly slashes are used for commenting.	// this comment will be ignored by //the program. And it will be //displayed in gray to help users write //their notes on the program
/* */	Different from double slash, a multi-line comment will allow you to write as much comments as you need between the slashes.	/* this comment will be ignored by the program. And it will be displayed by gray to help user write his notes on the program*/

dot.

Arduino IDE working process

Anything you write in the program will be read by the processor.

The program follows a certain algorithm during execution by going step by step through each comment/code written in the program.

If you add a code with a wrong syntax, your program won't work correctly or may stop the uploading process.

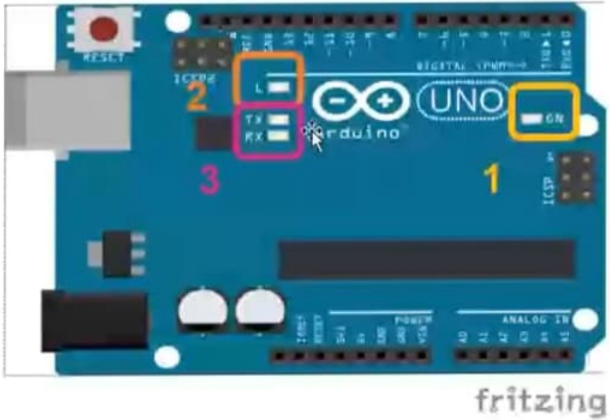
Let's find the LED

You can find four LED on an Arduino Uno. Each one of these LED has a task:

1- The ON LED will glow if you have your Arduino Turned ON

2- The L LED is a built-in LED that you can control

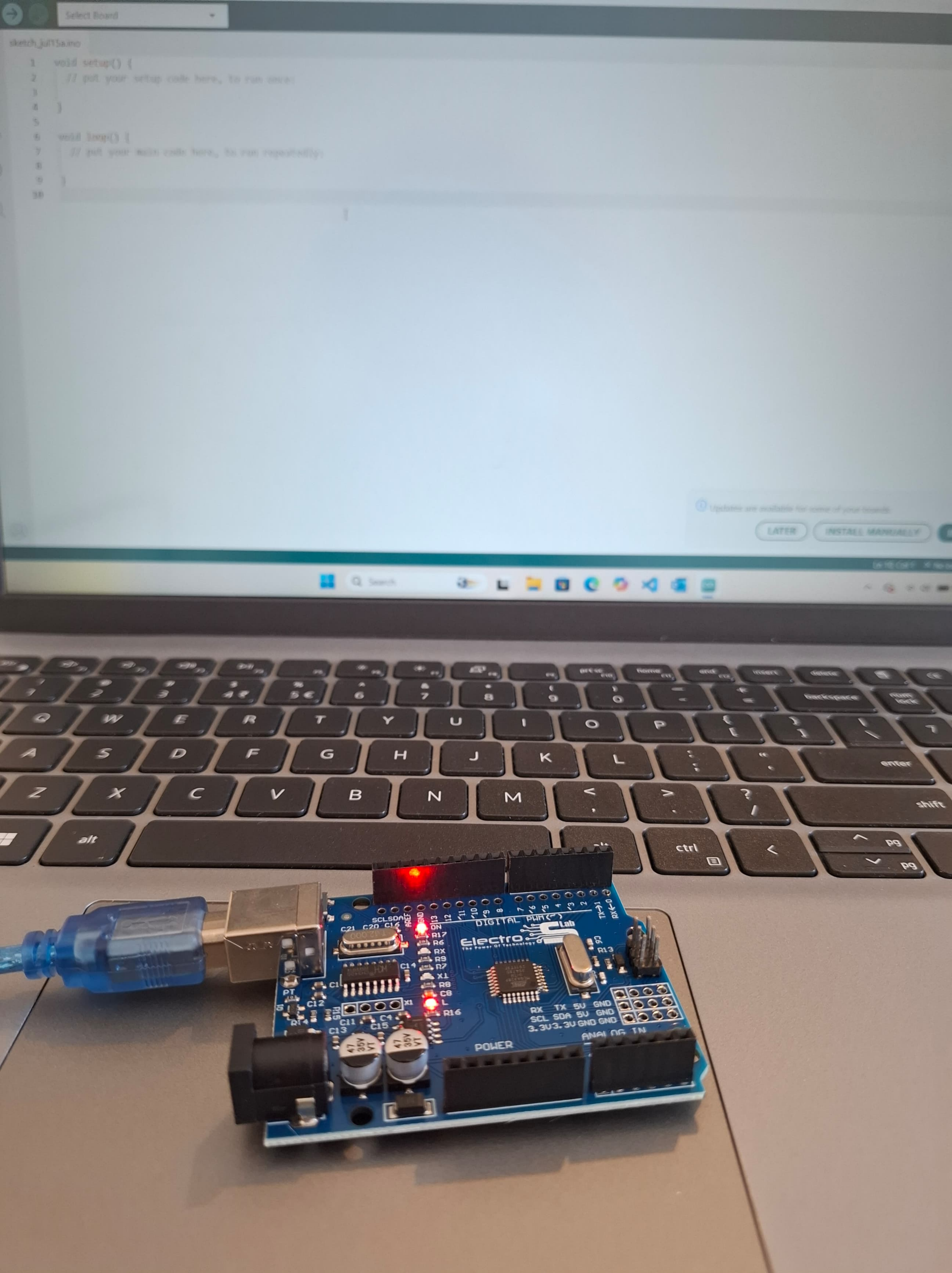
3- The TX and RX LED indicates Wi-Fi connection and programming download.



Activity 1: Turning on the built- in LED (L)

Let's try turning ON the L LED found on your Arduino board.

1. Open your Arduino IDE
2. Create A New File
3. Start writing, by defining the LED_BULTIN port as an output port. The LED_BUITLIN is a predefined variable that is connected internally with the breadboard. This should be done in the void Setup Function. Now this port can send a HIGH or LOW voltage signal.


Select Board
void setup() {
// put your setup code here, to run once:
}
void loop() {
// put your main code here, to run repeatedly:
}
LATER
INSTALL MANUALLY
DIGITAL PWM(~)
Electro
POWER
ANALOG IN
RX TX 5V GND
SCL SDA 5V GND
3.3V 3.3V GND GND

sketch_jul15a | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino UNO

sketch_jul15a.ino

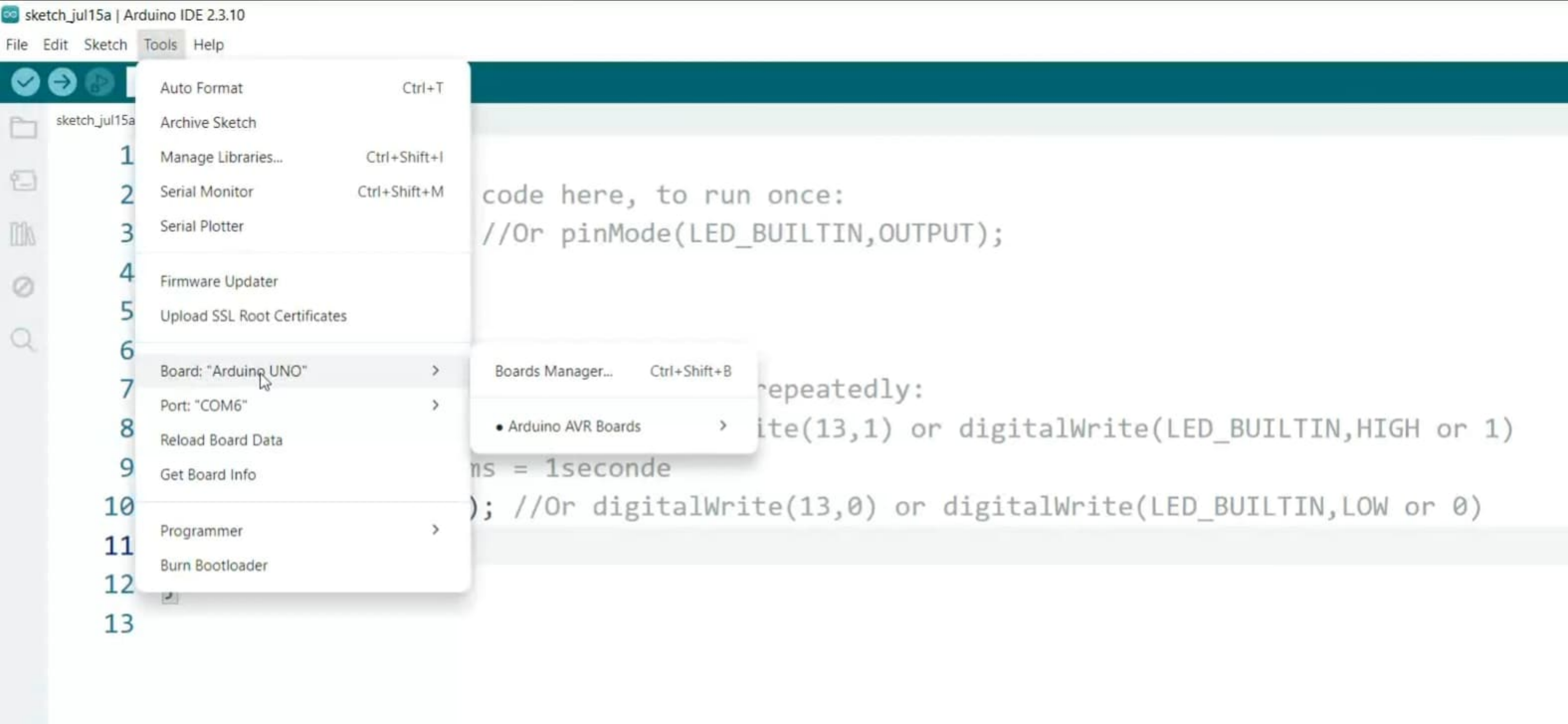
macro LED_BUILTIN

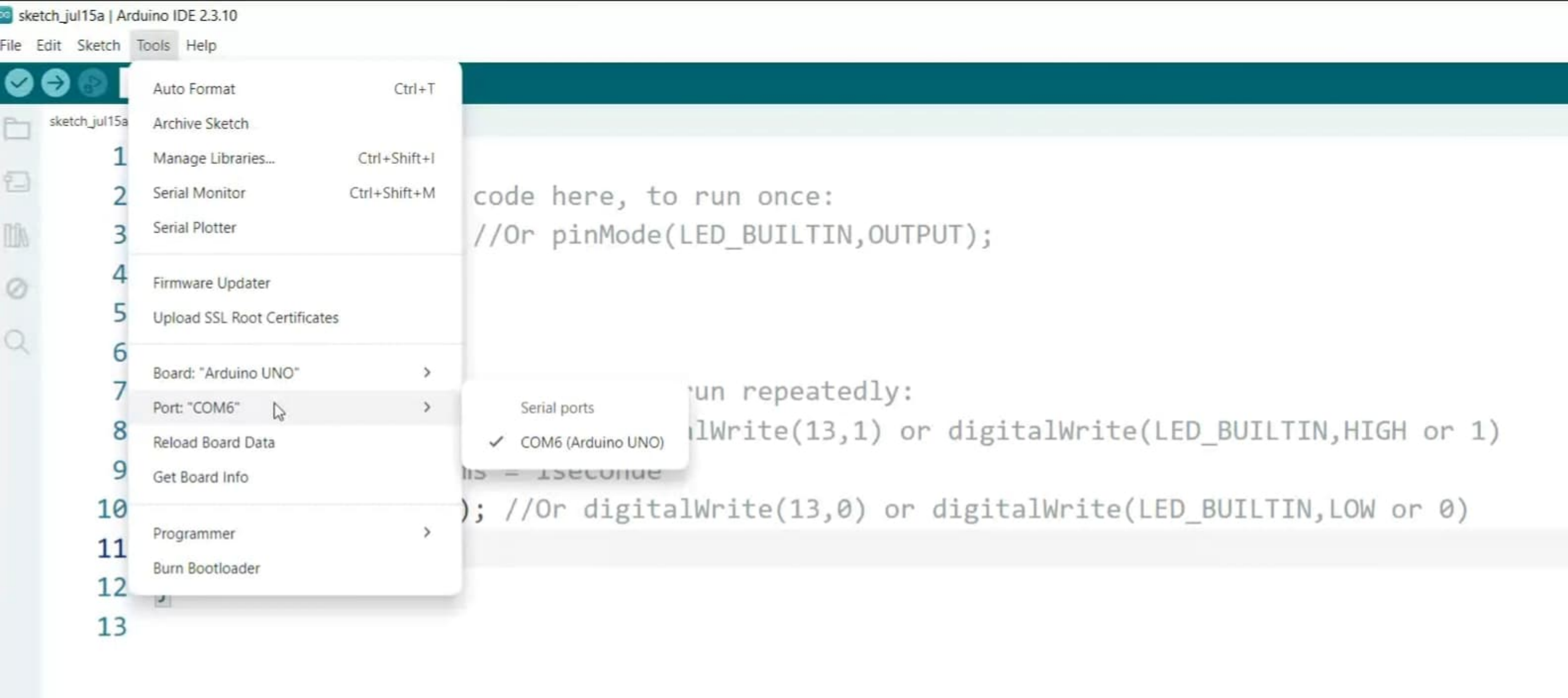
#define LED_BUILTIN 13

```
void set
  // put                                      to run once:
pinMode(LED_BUILTIN,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:

}

```

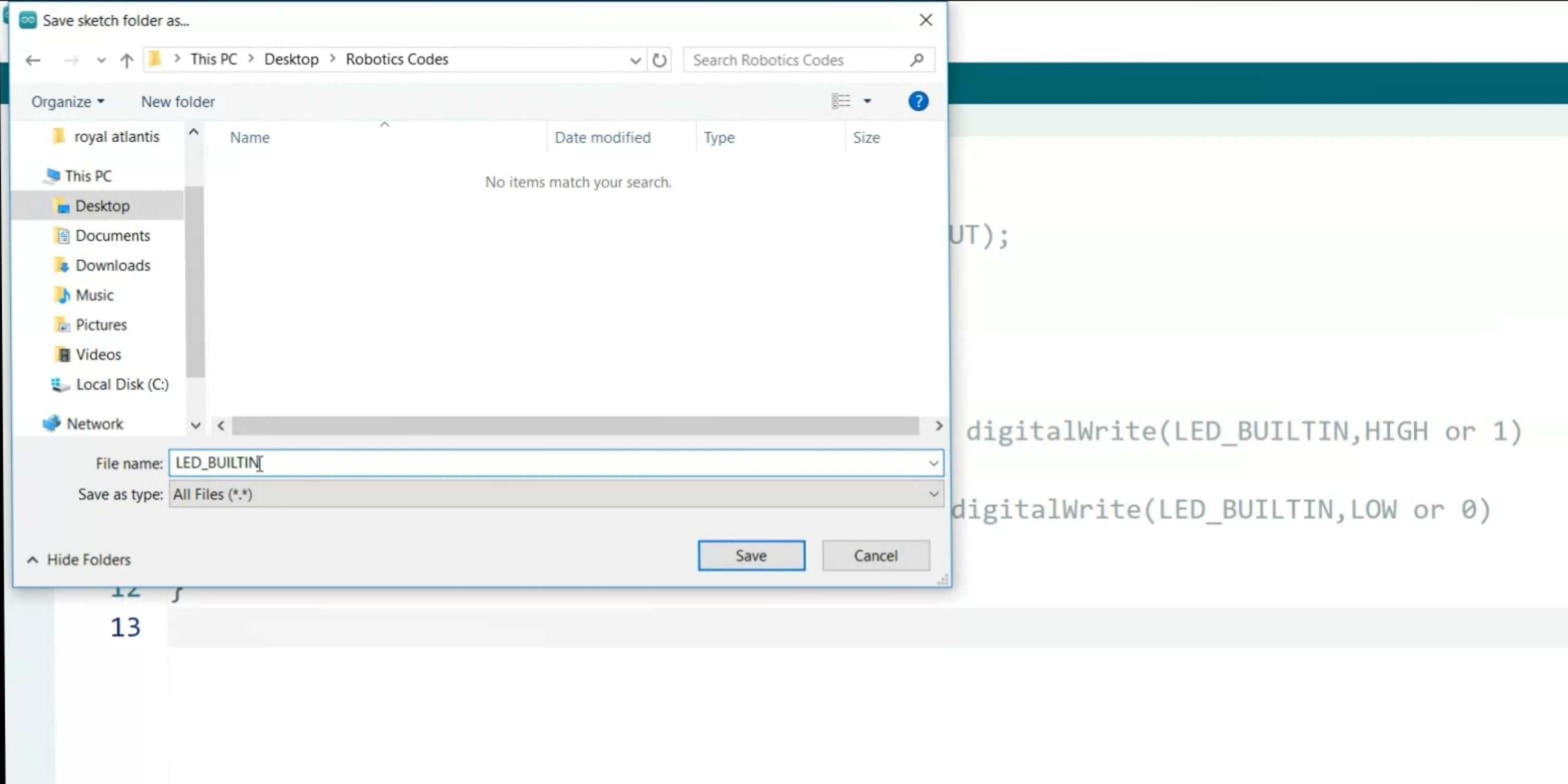
sketch_jul15a.ino

```
void setup() {
  // put your setup code here, to run once:
pinMode(13,OUTPUT); //Or pinMode(LED_BUILTIN,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
digitalWrite(13,HIGH); //Or digitalWrite(13,1) or digitalWrite(LED_BUILTIN,HIGH or 1)
delay(1000); //1000ms = 1seconde
digitalWrite(13,LOW); //Or digitalWrite(13,0) or digitalWrite(LED_BUILTIN,LOW or 0)
}

```


sketch_jul15a | Arduino IDE 2.3.10
File
Edit
Sketch
Tools
Help
sketch_jul15a
Auto Format Ctrl+T
Archive Sketch
Manage Libraries... Ctrl+Shift+I
Serial Monitor Ctrl+Shift+M
Serial Plotter
Firmware Updater
Upload SSL Root Certificates
Board: "Arduino UNO"
Port: "COM6"
Reload Board Data
Get Board Info
Programmer
Burn Bootloader
Boards Manager... Ctrl+Shift+B
Arduino AVR Boards
code here, to run once:
//Or pinMode(LED_BUILTIN,OUTPUT);
repeatedly:
ite(13,1) or digitalWrite(LED_BUILTIN,HIGH or 1)
ms = 1seconde
); //Or digitalWrite(13,0) or digitalWrite(LED_BUILTIN,LOW or 0)


sketch_jul15a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
sketch_jul15a
Auto Format Ctrl+T
Archive Sketch
Manage Libraries... Ctrl+Shift+I
Serial Monitor Ctrl+Shift+M
Serial Plotter
Firmware Updater
Upload SSL Root Certificates
Board: "Arduino UNO"
Port: "COM6"
Reload Board Data
Get Board Info
Programmer
Burn Bootloader
Serial ports
COM6 (Arduino UNO)
code here, to run once:
//Or pinMode(LED_BUILTIN,OUTPUT);
un repeatedly:
lWrite(13,1) or digitalWrite(LED_BUILTIN,HIGH or 1)
); //Or digitalWrite(13,0) or digitalWrite(LED_BUILTIN,LOW or 0)

sketch_jul15a | Arduino IDE 2.3.10
File
Edit
Sketch
Tools
Help
New Sketch Ctrl+N
New Cloud Sketch Alt+Ctrl+N
Open... Ctrl+O
Open Recent
Sketchbook
Examples
Close Ctrl+W
Save Ctrl+S
Save As... Ctrl+Shift+S
Preferences... Ctrl+Comma
Advanced
Quit Ctrl+Q
etup() {
ut your setup code here, to run once:
e(13,OUTPUT); //Or pinMode(LED_BUILTIN,OUTPUT);
oop() {
ut your main code here, to run repeatedly:
lWrite(13,HIGH); //Or digitalWrite(13,1) or digitalWrite(LED_BUILTIN,HIGH or 1)
1000); //1000ms = 1seconde
lWrite(13,LOW); //Or digitalWrite(13,0) or digitalWrite(LED_BUILTIN,LOW or 0)
1000);
12 }
13


Save sketch folder as...
This PC > Desktop > Robotics Codes
Search Robotics Codes
Organize
New folder
royal atlantis
This PC
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
Network
Name
Date modified
Type
Size
No items match your search.
File name: LED_BUILTIN
Save as type: All Files (*.*)
Hide Folders
Save
Cancel
UT);
digitalWrite(LED_BUILTIN,HIGH or 1)
digitalWrite(LED_BUILTIN,LOW or 0)
13


LED_BUILTIN | Arduino IDE 2.3.10
File
Edit
Sketch
Tools
Help
Undo Ctrl+Z
Redo Ctrl+Y
Cut Ctrl+X
Copy Ctrl+C
Copy for Forum (Markdown) Ctrl+Shift+C
Paste Ctrl+V
Select All Ctrl+A
Go to Line... Ctrl+L
Comment/Uncomment Ctrl+/
Increase Indent
Decrease Indent
Auto Format Ctrl+T
Replace in Files
Increase Font Size Ctrl+=
Decrease Font Size Ctrl+-
Find Ctrl+F
Find Next Ctrl+G
Find Previous Ctrl+Shift+G
Use Selection for Find Ctrl+E
{
setup code here, to run once:
TPUT); //Or pinMode(LED_BUILTIN,OUTPUT);
main code here, to run repeatedly:
13,HIGH); //Or digitalWrite(13,1) or digitalWrite(LED_BUILTIN,HIGH or 1)
//1000ms = 1seconde
13,LOW); //Or digitalWrite(13,0) or digitalWrite(LED_BUILTIN,LOW or 0)

LED_BUILTIN | Arduino IDE 2.3.10

ile Edit Sketch Tools Help

Arduino UNO

LED_BUILTIN.ino

```
void setup() {
  // put your setup code here, to run once:
pinMode(13,OUTPUT); //Or pinMode(LED_BUILTIN,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
digitalWrite(13,HIGH); //Or digitalWrite(13,1) or digitalWrite(LED_BUILTIN,HIGH or 1)
delay(1000); //1000ms = 1seconde
digitalWrite(13,LOW); //Or digitalWrite(13,0) or digitalWrite(LED_BUILTIN,LOW or 0)
delay(1000);
}
```

LED_BUILTIN | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino UNO

LED_BUILTIN.ino

```
void setup() {
  // put your setup code here, to run once:
pinMode(13,OUTPUT); //Or pinMode(LED_BUILTIN,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
digitalWrite(13,HIGH); //Or digitalWrite(13,1) or digitalWrite(LED_BUILTIN,HIGH or 1)
delay(1000); //1000ms = 1seconde
digitalWrite(13,LOW); //Or digitalWrite(13,0) or digitalWrite(LED_BUILTIN,LOW or 0)
delay(1000);

exit(0);
}

```

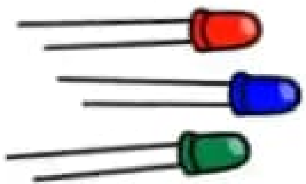
Output

```
Sketch uses 924 bytes (2%) of program storage space. Maximum is 32256 bytes.
Global variables use 9 bytes (0%) of dynamic memory, leaving 2039 bytes for local variables. Maximum is 2048 bytes.
Warning: linked C library does not conform to C99; avrdude may not work as expected
```

What is a LED?

In the simplest terms, a light-emitting diode (LED) is a semiconductor device that emits light when an electric current is passed through it.

The LED can be used in robots for sending visual messages like if it is going forward (blinking green), Steady (steady green) or in case of an error (bright red).



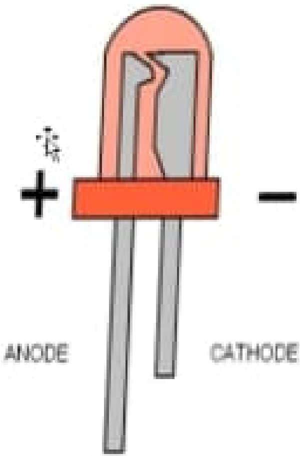
LEDS (light emitting diodes) emit light when a voltage is applied.

dot.

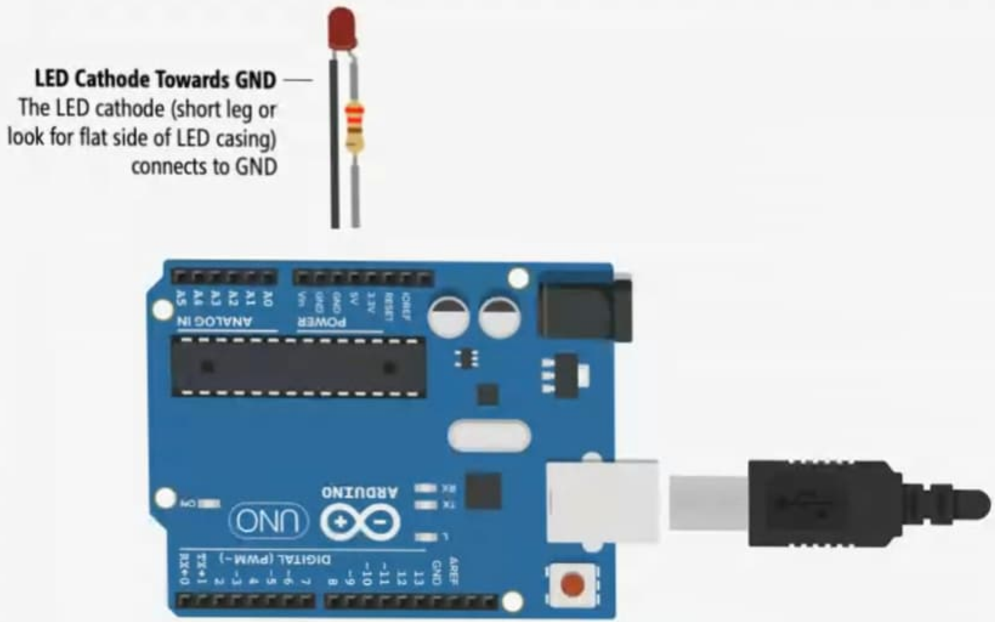
LED characteristics

The LED has some characteristics :

1. Each LED requires a fixed amount of voltage to work, the LED in your kit needs 5 VOLT
2. The LED has two legs: an anode and a cathode.
3. The anode is the long leg. It should go to the positive port or VCC port on the Arduino or to a digital pin that is in OUTPUT mode.
4. The cathode is the short leg. It should go to the negative or GRD port on the Arduino.



LED characteristics



Resistor

- A passive electrical component with two terminals that are used for either limiting or regulating the flow of electric current in electrical circuits and to lower the voltage in any particular portion of the circuit.

  - This is necessary because a pin can encounter an electric flow of maximum 40 mA (standard in Arduino UNO boards but not a standard in all boards).

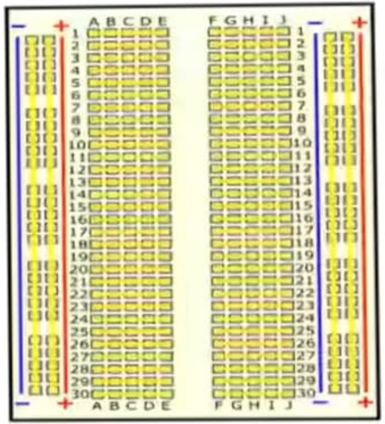
  - If you want to connect a LED or a push button directly to the Arduino, you should add a resistor.

  - Each resistor is identified by the colored lines.

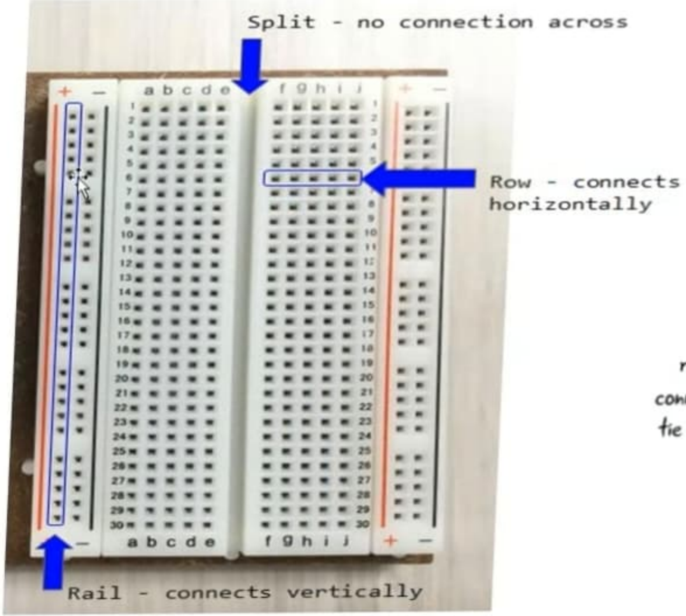
dot. LEBANON

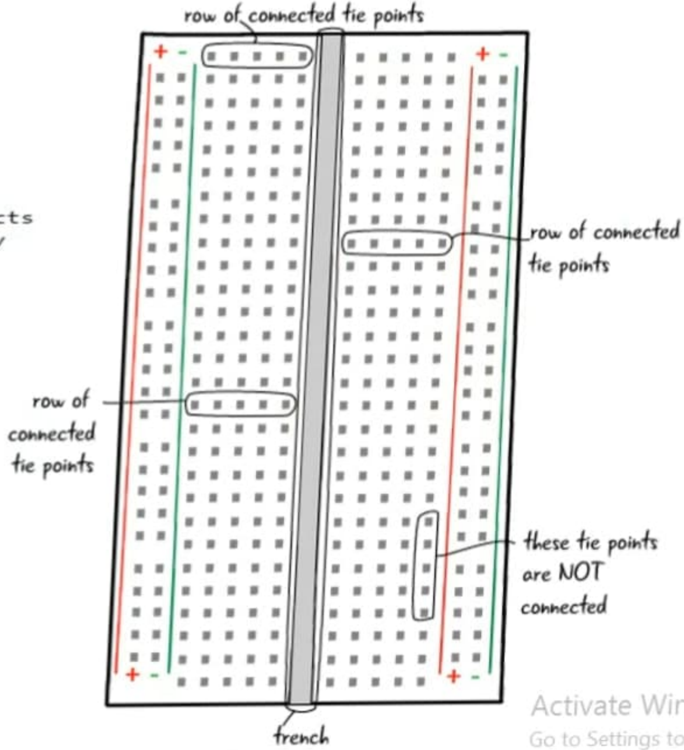
Breadboard

How are each column or row linked together?



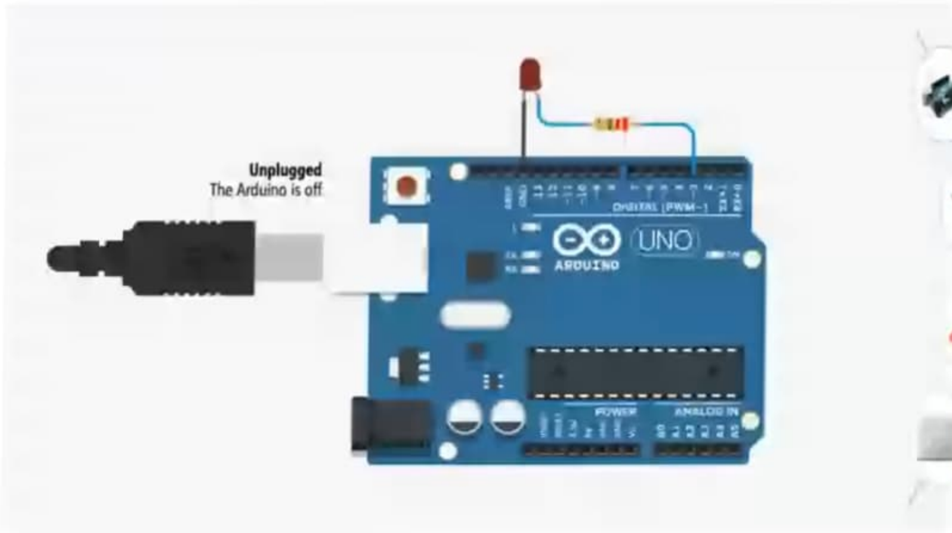
Breadboard


Split - no connection across
a b c d e
f g h i j
Row - connects horizontally
Rail - connects vertically


row of connected tie points
row of connected tie points
row of connected tie points
these tie points are NOT connected
trench

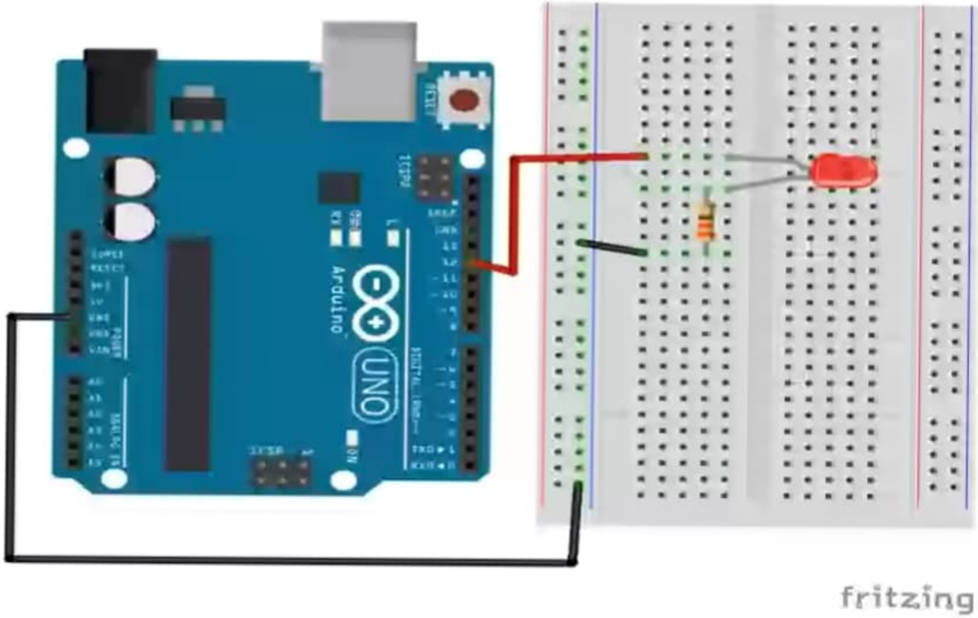
Control an external LED

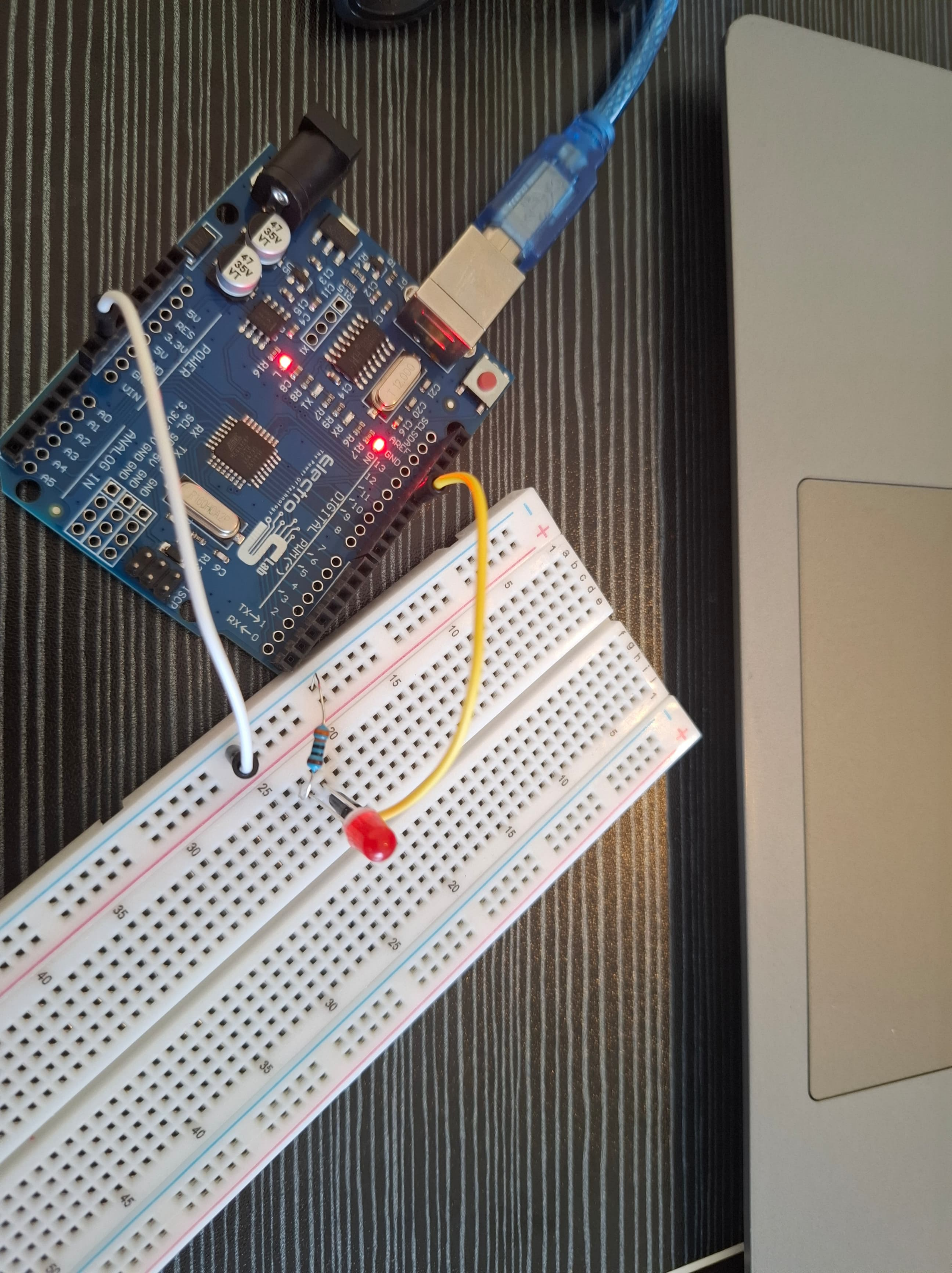
Required Elements:

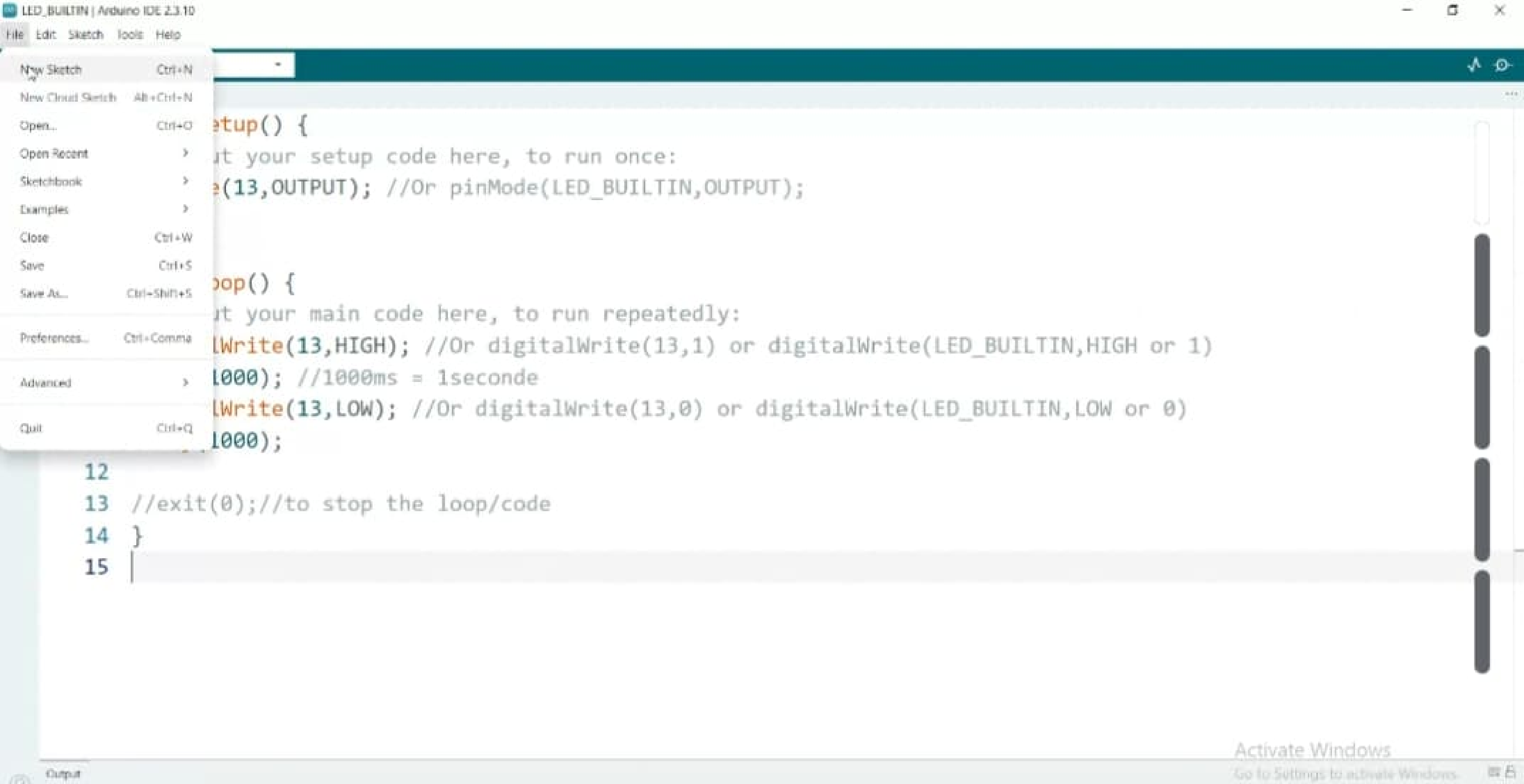


- Arduino
- Resistor 330 Ω
- Red LED
- Breadboard

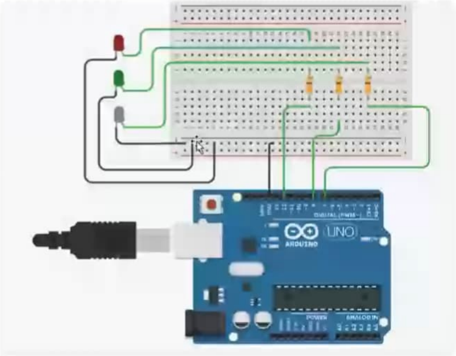
Board connection

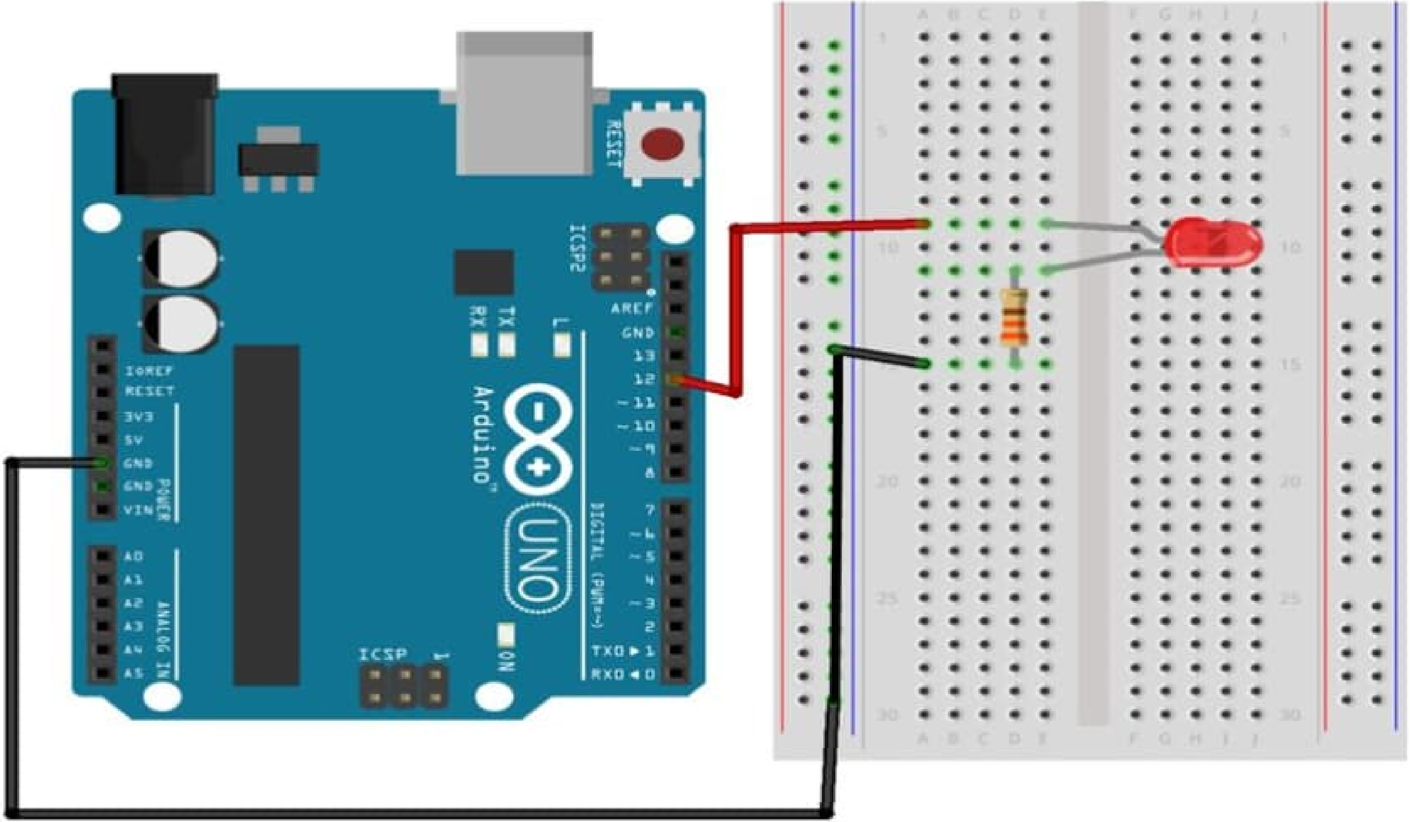


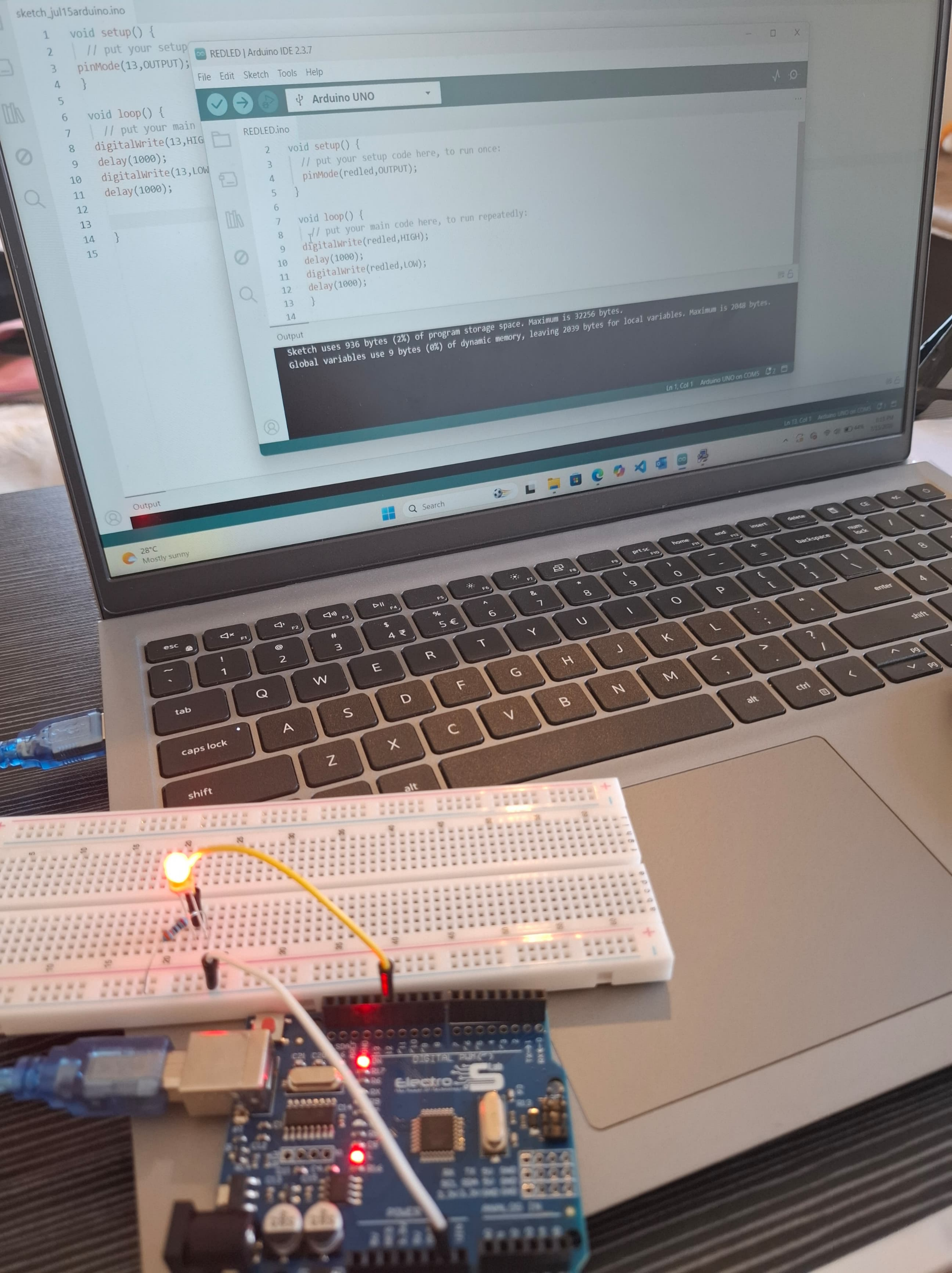

electro
DIGITAL PWM
POWER
ANALOG IN


LED_BUILTIN | Arduino IDE 2.3.10
File Edit Sketch Tools Help
New Sketch Ctrl+N
New Cloud Sketch Alt+Ctrl+N
Open... Ctrl+O
Open Recent
Sketchbook
Examples
Close Ctrl+W
Save Ctrl+S
Save As... Ctrl+Shift+S
Preferences... Ctrl+Comma
Advanced
Quit Ctrl+Q
etup() {
ut your setup code here, to run once:
e(13,OUTPUT); //Or pinMode(LED_BUILTIN,OUTPUT);
oop() {
ut your main code here, to run repeatedly:
lWrite(13,HIGH); //Or digitalWrite(13,1) or digitalWrite(LED_BUILTIN,HIGH or 1)
1000); //1000ms = 1seconde
lWrite(13,LOW); //Or digitalWrite(13,0) or digitalWrite(LED_BUILTIN,LOW or 0)
1000);
12
13 //exit(0);//to stop the loop/code
14 }
15
Activate Windows
Go to Settings to activate Windows.
Output

Board connection




RESET
ICSP2
RX
TX
L
AREF
GND
13
12
~11
~10
~9
8
IOREF
RESET
3V3
5V
GND
GND
VIN
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
Arduino
UNO
DIGITAL (PWM=~)
7
~6
~5
4
~3
2
TX0 ▶ 1
RX0 ◀ 0
ICSP
1
ON


sketch_jul15arduino.ino
void setup() {
// put your setup
pinMode(13,OUTPUT);
}
void loop() {
// put your main
digitalWrite(13,HIG
delay(1000);
digitalWrite(13,LOW
delay(1000);
}
REDLED | Arduino IDE 2.3.7
File Edit Sketch Tools Help
Arduino UNO
REDLED.ino
void setup() {
// put your setup code here, to run once:
pinMode(redled,OUTPUT);
}
void loop() {
// put your main code here, to run repeatedly:
digitalWrite(redled,HIGH);
delay(1000);
digitalWrite(redled,LOW);
delay(1000);
}
Output
Sketch uses 936 bytes (2%) of program storage space. Maximum is 32256 bytes.
Global variables use 9 bytes (0%) of dynamic memory, leaving 2039 bytes for local variables. Maximum is 2048 bytes.
Ln 1, Col 1 Arduino UNO on COM5
Search
28°C
Mostly sunny
DIGITAL PWM(~)